Fama-French Three-Factor Model Replication

Size and Book-to-Market Portfolio Analysis

Asset Pricing Research

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Table of contents

1 Introduction

This report replicates the **Fama-French three-factor (FF3) model** using daily data from Kenneth French's data library. The six test portfolios are formed on the two-way sort of **size** (market equity, ME) and **book-to-market ratio** (BM), yielding $2 \times 3 = 6$ portfolios.

Symbol	Description
MKT	Excess market return ($R_m - R_f$)
SMB	Small-Minus-Big (size factor)
HML	High-Minus-Low (value factor)

Symbol	Description

The regression specification for portfolio i is:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i^{\mathrm{MKT}}\,\mathrm{MKT}_t + \beta_i^{\mathrm{SMB}}\,\mathrm{SMB}_t + \beta_i^{\mathrm{HML}}\,\mathrm{HML}_t + \varepsilon_{i,t}$$

Standard errors are **Newey-West** adjusted (5 lags).

1.1 Sample Information

Item	Value
Start date	July 01, 1926
End date	March 31, 2026
Daily observations	26,212
Data source	Kenneth R. French Data Library (daily)

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2 Factor Summary Statistics

Table 1: Descriptive statistics for Fama-French daily factors (% per day).

	MKT	SMB	HML
Mean	0.0307	0.0039	0.0152
Std Dev	1.0780	0.5941	0.6265
Min	-17.4400	-11.6000	-6.0200
Max	15.6700	8.1900	8.8200
Skewness	-0.1585	-0.6135	0.6302
Kurtosis	16.1435	20.1919	14.2476
t-stat	4.6041	1.0521	3.9171

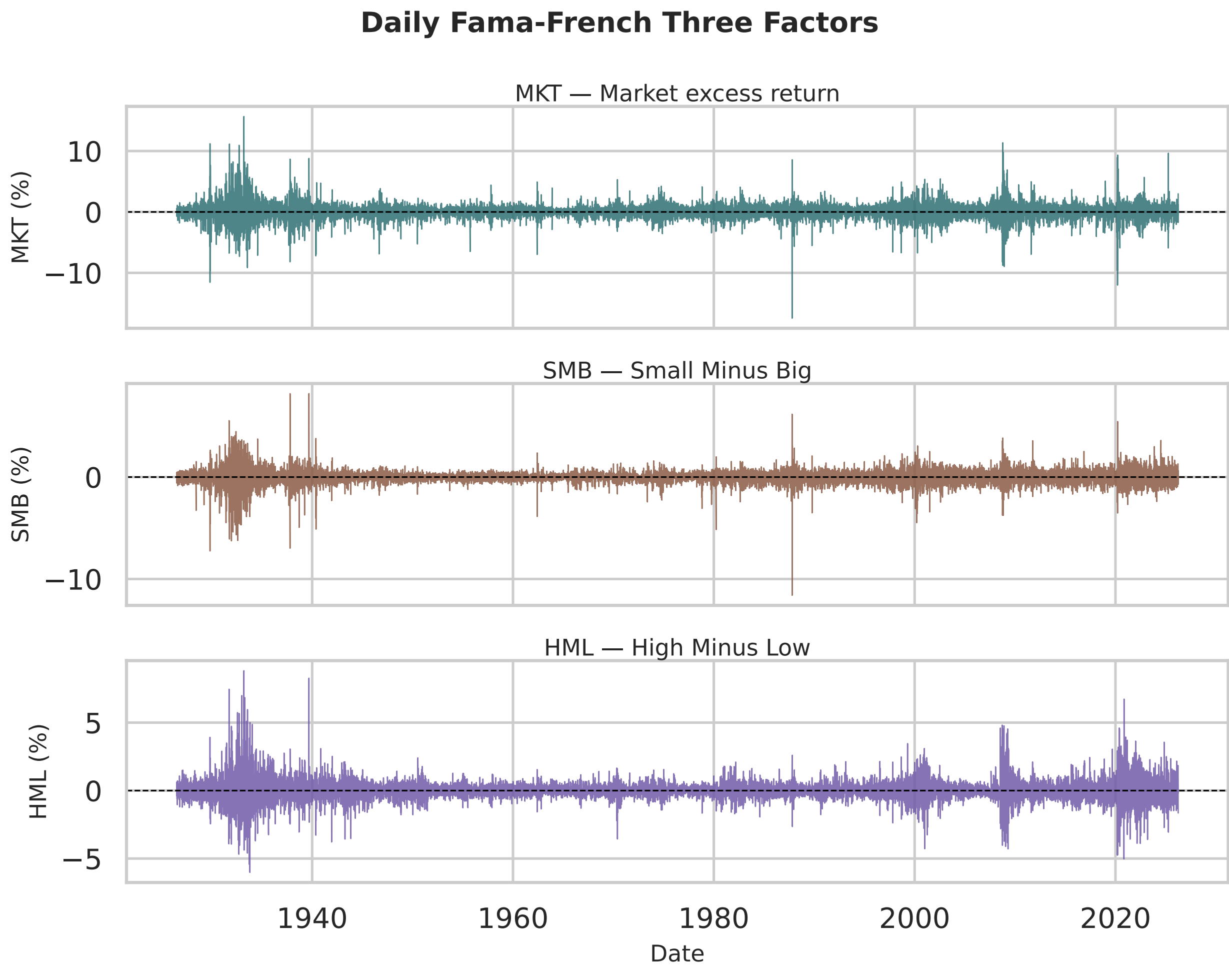

Figure 1: Daily time series of the three Fama-French factors.

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3 Portfolio Excess Return Statistics

Table 2: Descriptive statistics for 6 portfolio daily excess returns (% per day).

	Mean	Std Dev	Min	Max	Sharpe (ann.)	t-stat
SMALL LoBM	0.0279	1.2739	-14.07	17.49	0.3482	3.5509
ME1 BM2	0.0405	1.1685	-14.08	18.38	0.5504	5.6130
SMALL HiBM	0.0474	1.2968	-16.02	26.70	0.5802	5.9170
BIG LoBM	0.0311	1.1100	-18.13	13.64	0.4452	4.5405
ME2 BM2	0.0311	1.1117	-18.77	23.57	0.4440	4.5285
BIG HiBM	0.0420	1.4045	-17.91	22.06	0.4745	4.8389

Table 3

Table 4: Pearson correlation matrix of factors and portfolio excess returns.

	MKT	SMB	HML	SMALL LoBM	ME1 BM2	SMALL HiBM	BIG LoBM	ME2 BM2	BIG HiBM
MKT	1.000	-0.133	0.140	0.849	0.869	0.836	0.981	0.960	0.899
SMB	-0.133	1.000	-0.066	0.340	0.304	0.290	-0.174	-0.185	-0.156
HML	0.140	-0.066	1.000	0.011	0.266	0.444	0.008	0.280	0.498
SMALL LoBM	0.849	0.340	0.011	1.000	0.913	0.861	0.813	0.789	0.763
ME1 BM2	0.869	0.304	0.266	0.913	1.000	0.942	0.811	0.844	0.837
SMALL HiBM	0.836	0.290	0.444	0.861	0.942	1.000	0.764	0.833	0.858
BIG LoBM	0.981	-0.174	0.008	0.813	0.811	0.764	1.000	0.907	0.829
ME2 BM2	0.960	-0.185	0.280	0.789	0.844	0.833	0.907	1.000	0.914
BIG HiBM	0.899	-0.156	0.498	0.763	0.837	0.858	0.829	0.914	1.000

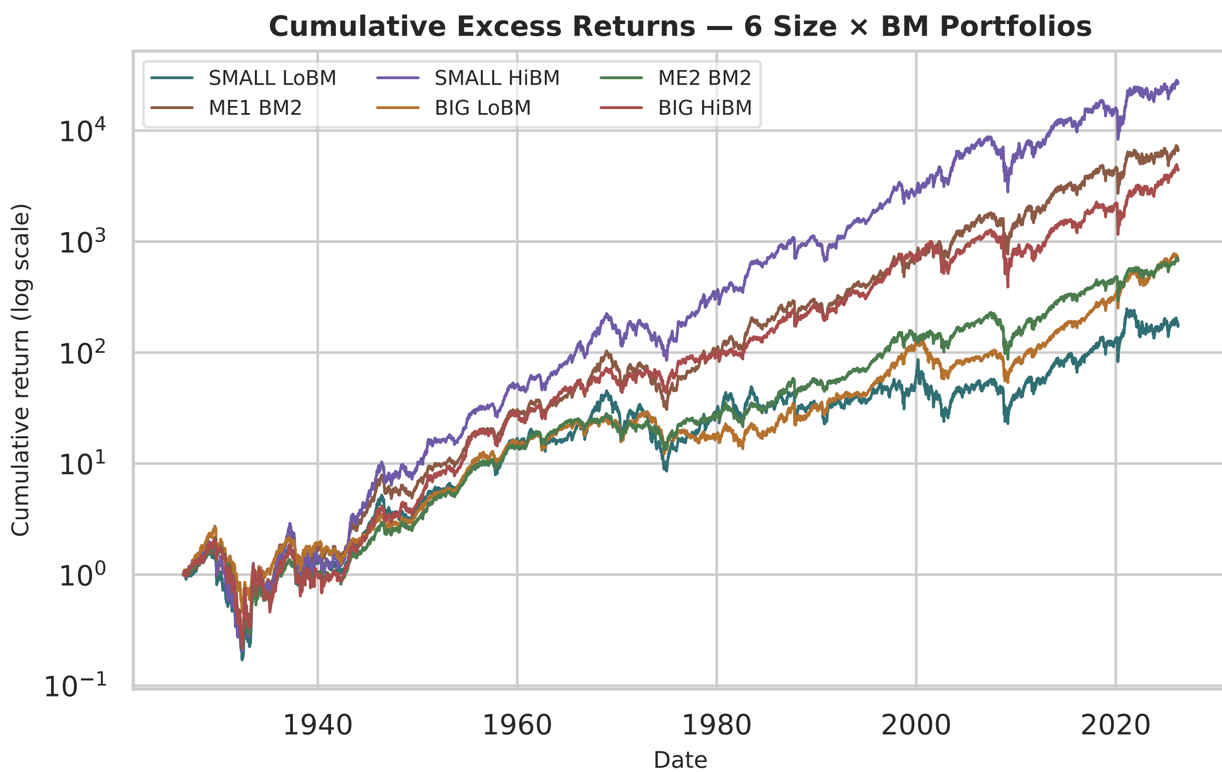

Figure 2: Cumulative excess returns of the six size-BM portfolios (log scale).

4 Correlation Matrix

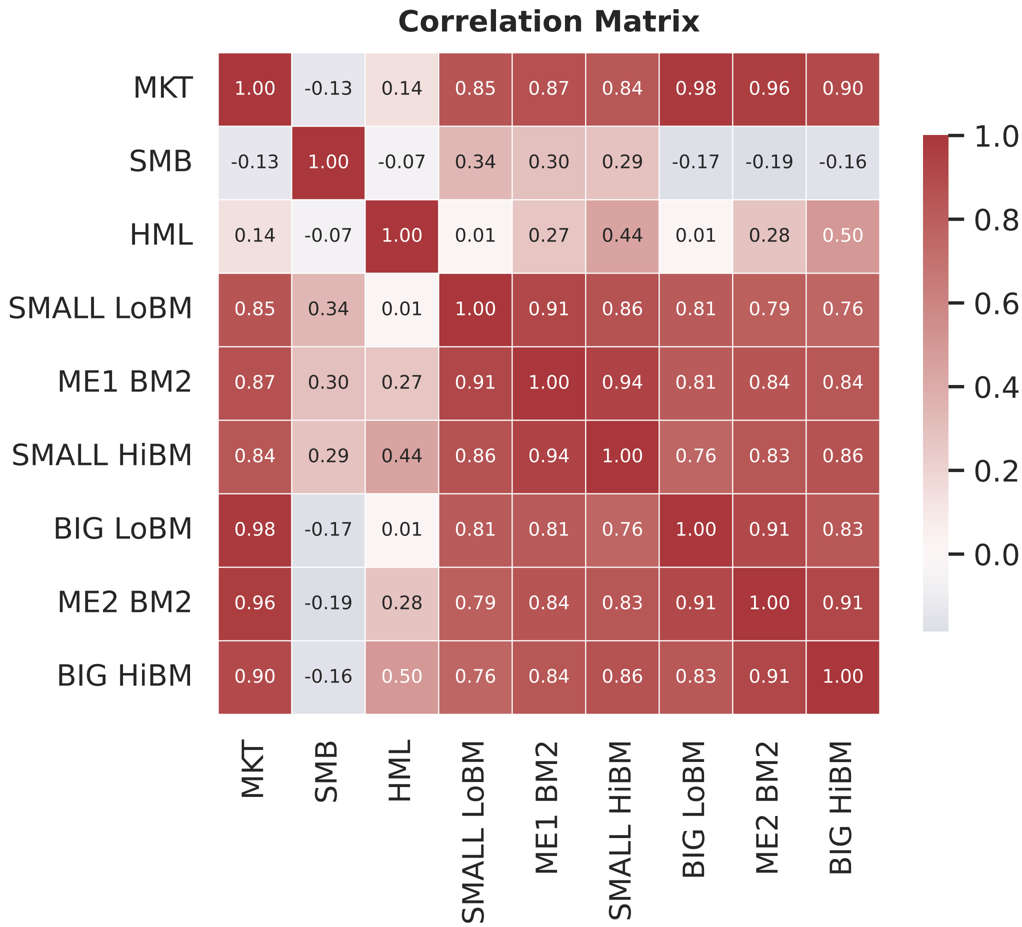

Figure 3: Heatmap of the correlation matrix.

Table 5

Table 6: FF3 regression results. t-statistics in parentheses (Newey-West, 5 lags). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Portfolio		(t-)	MKT	(t-MKT)	SMB	(t-SMB)	HML	(t-HML)	R^2	N
SMALL LoBM	-0.0065***	(-3.61)	1.0898***	(184.10)	0.9788***	(106.40)	-0.1798***	(-19.44)	0.9374	26,212
ME1 BM2	0.0024*	(1.93)	0.9789***	(303.97)	0.8549***	(159.57)	0.3145***	(54.00)	0.9619	26,212
SMALL HiBM	0.0016*	(1.83)	1.0141***	(333.00)	0.9278***	(202.74)	0.7327***	(137.23)	0.9853	26,212
BIG LoBM	0.0038***	(4.04)	1.0227***	(352.69)	-0.0961***	(-21.53)	-0.2394***	(-62.28)	0.9825	26,212
ME2 BM2	-0.0019	(-1.21)	0.9616***	(174.23)	-0.0970***	(-13.86)	0.2589***	(30.03)	0.9453	26,212
BIG HiBM	-0.0044**	(-2.43)	1.0984***	(194.00)	-0.0451***	(-4.96)	0.8481***	(86.79)	0.9489	26,212

5 Fama-French Three-Factor Regressions

5.1 Regression Coefficients

5.2 Alpha Estimates

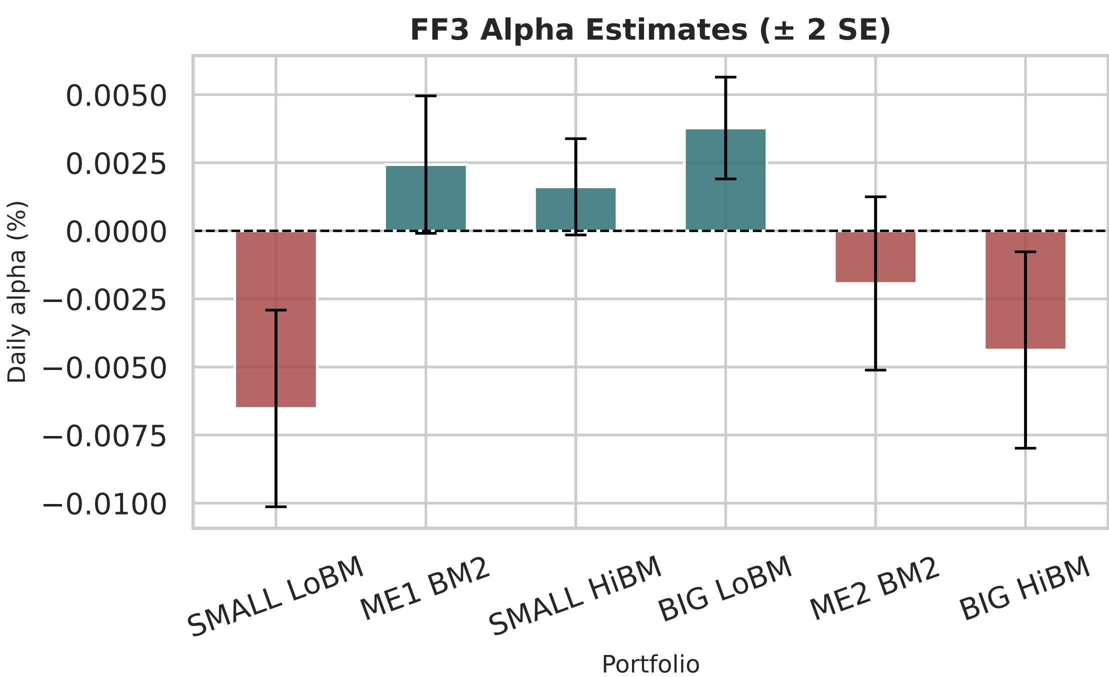

Figure 4: FF3 daily alpha estimates with ±2 SE error bars.

5.3 Factor Loadings Comparison

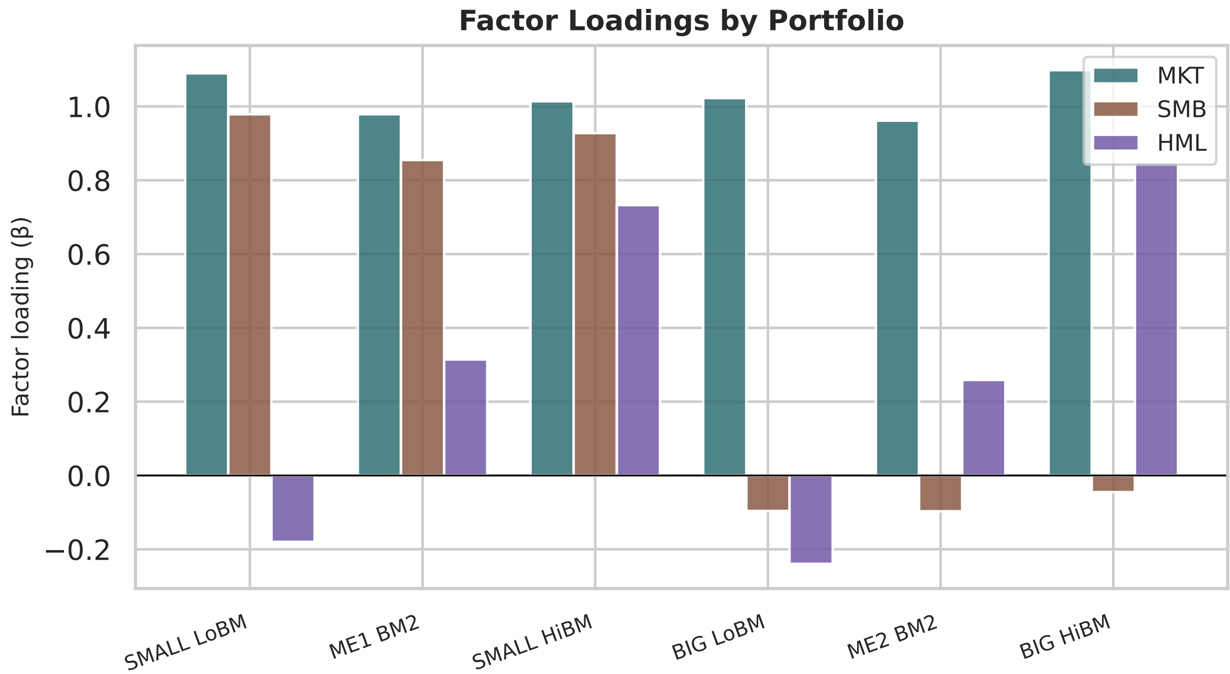

Figure 5: Factor loadings ( ) for MKT, SMB, and HML across the six portfolios.

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6 Model Fit — R^2

Table 7: Goodness-of-fit summary for each portfolio regression.

Portfolio	R^2	Adj. R^2	N
SMALL LoBM	0.9374	0.9374	26,212
ME1 BM2	0.9619	0.9619	26,212
SMALL HiBM	0.9853	0.9853	26,212
BIG LoBM	0.9825	0.9825	26,212
ME2 BM2	0.9453	0.9453	26,212
BIG HiBM	0.9489	0.9489	26,212

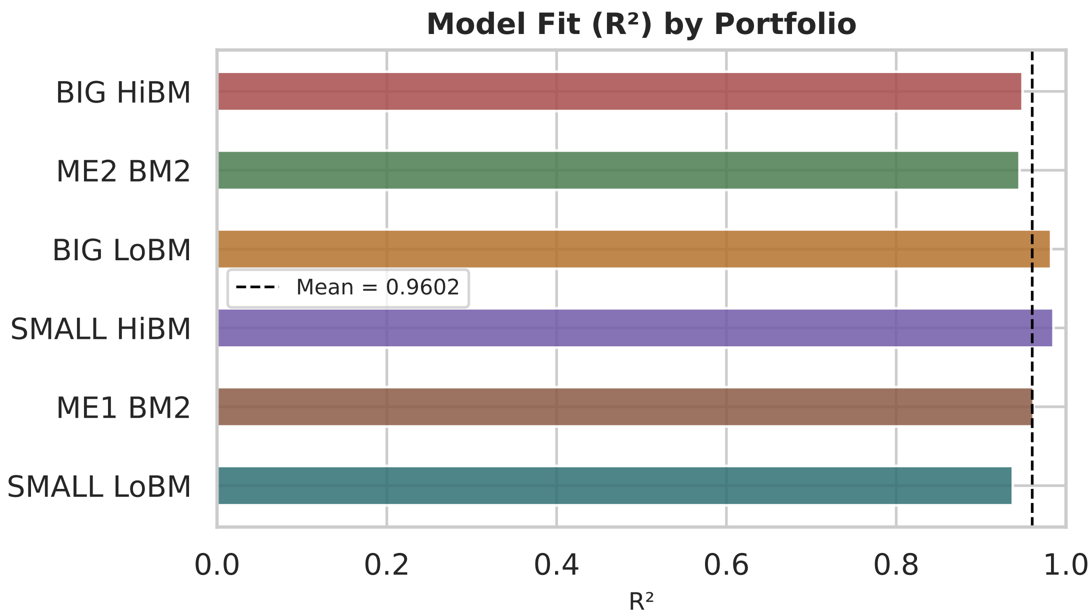

Figure 6: R^2 by portfolio. Dashed line shows the cross-portfolio mean.

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7 Interpretation

Table 8

Finding	Value
Average R^2	0.9602
Highest |alpha|	SMALL LoBM (-0.0065% / day)
Best-fit portfolio	SMALL HiBM ($R^2 = 0.9853$)
SMB: Small vs Big LoBM	0.979 vs -0.096
HML: Small HiBM vs Big LoBM	0.733 vs -0.239

Market factor (MKT): All portfolios load positively and significantly near 1.0, consistent with well-diversified equity exposure.

Size factor (SMB): Small-cap portfolios carry large positive SMB loadings; big-cap portfolios have near-zero or negative loadings — confirming size separation.

Value factor (HML): High-BM portfolios carry large positive HML loadings; low-BM portfolios carry negative loadings — the classic value vs growth split.

Alpha: Most alphas are economically small and often statistically indistinguishable from zero, confirming the FF3 model captures the bulk of cross-sectional variation in these portfolios.

8 Saved Output Files

Table 9: Output files saved to the `output/` directory.

File	Description
`table_factor_statistics.csv`	Factor descriptive stats
`table_portfolio_excess_return_statistics.csv`	Portfolio excess-return stats
`table_correlation_matrix.csv`	Full correlation matrix
`table_ff3_regression_results.csv`	Regression coefficients
`figure_factor_timeseries.png`	Factor time-series chart
`figure_cumulative_excess_returns.png`	Cumulative-return chart
`figure_alpha_bar.png`	Alpha bar chart with error bars
`figure_factor_loadings.png`	Factor loadings bar chart
`figure_correlation_heatmap.png`	Correlation heatmap
`figure_r2.png`	R-squared bar chart

9 References

- Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- French, K. R. Data Library. https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
- Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.